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Visual perception based deep learning transformers for classifying paintings and photographs through feature extraction

Liu Yu

The application of computer vision techniques that have been used in various domains and artwork is not an exception. The latest models of deep learning are applied for processing and classification of digital images to analyze textures, color compositions, and lighting patterns across artistic and real-world imagery. In this research work, we aim to classify the given images to identify whether they are human artwork of paintings or captured photos. We apply state-of-the-art Vision Transformer (ViT) architecture to classify artistic images, achieving a classification accuracy of 95% outperforming existing relevant in the relevant literature. To validate and benchmark the approach, comprehensive experiments are carried out on the standard dataset using DenseNet, Convolutional Neural Networks (CNN), and Visual Geometry Group (VGG19) models. The results highlight the superior performance of ViT in capturing complex visual features, such as texture variation and compositional details. Grad-CAM further enhances the interpretability of the model by highlighting the specific visual regions influencing the model's decisions, confirming that ViT able to capture meaningful artistic attributes such as brushstroke patterns, smoothness, and illumination gradients. This combination of high performance and transparency analysis of proposed model ensures a reliable and explainable solution for automated artwork classification.

Keywords Vision transformer, Attention mechanism, Transformer block, Deep learning, Human-Created paintings, Image classification, Artistic style recognition, Feature extraction, Visual art analysis

Art and photography have played a fundamental role in human civilization, serving as powerful mediums of expression, documentation, and innovation. From ancient cave paintings to digital photography, these creative forms have influenced culture, history, and even economies[1]. Art, in its many forms, reflects human emotion, thought, and societal evolution, while photography serves as a bridge between reality and perception, capturing moments with precision and depth[2]. Paintings and photographs differ fundamentally in their visual structure, texture, and composition. Paintings typically exhibit brushstroke patterns, stylized color distributions[3], and abstract or interpretive elements[4], while photographs capture realistic details, sharp textures, and natural lighting variations[5]. These inherent differences create distinct feature distributions, which the model must learn to differentiate during training[6].

Together, they shape narratives, evoke emotions[7], and inspire movements across disciplines. The intersection of technology and art has ushered in a new era of creativity, where Artificial Intelligence (AI) is transforming the way to produce and perceive visual art[8]. From generative adversarial networks (GANs) creating hyper-realistic images to machine learning algorithms mimicking the style of renowned painters, AI has blurred the boundaries between human-made and machine-generated art[9]. Classification of art, especially in terms of artistic styles, is an important and rapidly expanding discipline in the art industry as well as in the technology industries[10]. The adoption of deep learning in the field of art analysis has experienced a remarkable rate of increase over the past few years, estimated to be above 30% per year, which explains the growing relevance behind it[11]. This trend is not merely motivated by the abstract worth of aesthetics but also by the real-life usage like authentication, restoration, and digital archiving[12]. With artworks shifting to electronic formats, it is highly important that the

School of Design and Fine Arts, Qingdao Huanghai University, Qingdao 266555, Shandong, China. email: LiuYu6464632@163.com

AI can effectively identify and classify artistic styles[13]. With its sophisticated pattern recognition features, deep learning becomes central to this process, allowing a more sophisticated perception and interaction with art[14].

Art and artificial intelligence not only interact to make art forms more engaging and comprehensible but also aid in key tasks of preservation and distribution of cultural heritage[15,16]. When properly implemented, these models can tell the difference between painting and real photographs for checking their authenticity and aesthetics-related features[17]. The creation of the detection technique is also a difficult issue since the works created by AI are increasingly similar to the traditional ones in terms of their authenticity in terms of the textures and brushwork and composition patterns[18].

This study focuses on understanding the growing application of AI making art by creating ways of validating conventional artistic processes coexisting with supplying AI manipulation. Galleries and art historians along with collectors are constantly struggling to establish not only the provenance of a work but also the way its creator created it. The paper deals with the problem of artistic classification to establish a reliable model that can distinguish between automated image making and genuine artwork to guard the integrity of the art medium. The problem addressed in this study is a binary image classification problem. A comparison of AI-generated pictures with traditional paintings in the context of their main aspects presents unique knowledge about artistic communication and the machine-based interpretation of creative behavior. The classification in AI-generated and real paintings is more distinguishable and better ensures the preservation of the integrity of the art market. Technologically speaking the study shows the strengths and weaknesses that the current artificial intelligence systems have when imitating human creative works. Two basic implications of AI technology are highlighted in the research since it alters the cultural landscapes and establishes discussions of evolving concepts of original art and creative activity. The inquiry that streamlines deep learning procedures comes to numerous breakthroughs in the scope of calculations as well as increasing our community discourse concerning the way artificial intelligence functions in the artistic world and the social systems. Our research contributions are as follows:

- Proposed a ViT-based model integrating self-attention mechanisms and transformer blocks to capture complex visual dependencies in artistic images, enhancing the model's ability to discriminate subtle differences between AI-generated and human-made artworks.
- Demonstrated that the ViT model achieved the highest classification accuracy of 95%, outperforming all baseline models and confirming the superiority of transformer-based architectures for detailed visual art recognition.
- Conducted comprehensive statistical significance testing, including ANOVA, t-test, Wilcoxon signed-rank test, Chi-square test, and DeLong AUC test, to rigorously validate the performance differences between the proposed model and baseline architectures, thereby ensuring the reliability, and statistical validity of the results.

The remainder of this paper is structured as follows: Sect. 2 reviews the existing literature on interactive media and its relevance to this study. The paper outlines methodological research details in Sect. 3 through description of experimental procedures and classification model structures. The research findings introduced in Sect. 4 analyze the study data with a thorough exploration of obtained results. Section 5 ends the paper by reviewing essential findings and specifying research guidelines for coming investigations.

Literature review

Recent research shows that deep learning classifiers, if trained on large amounts of data, can differentiate images with a good degree of accuracy[19]. Popular convolutional neural networks (CNNs) also compute an empirical decision rule that is then used for classification: for instance, ResNet CNN trained on a mixed real vs. AI image dataset (538 AI vs. 435 real) yielding 83% test accuracy[20]. Similarly, Yolov7[21], CNN based on DenseNet-121 using a large, labeled set of natural as well as GAN-generated images with 93.4% accuracy[22]. More recently, it has been applied widely to different research fields, that further fine-tuned ViT-B/ViT-L on interior design images (1000 real vs. 1000 generated) and achieves 86.3% accuracy[23].

Another effective approach is feature engineering, designed manually crafted frequency-domain features from StyleGAN-prodigy datasets and train classic classifiers, they achieve accuracy ≥ 87% and 70% with an SVM-RBF on a subset of the transforms[24]. Furthermore, using error-level analysis (ELA) and PRNU to estimate image noise features and to feed them to CNN for training; the CNN trained via ELA achieved 88% accuracy (and 99% when using both PRNU and ELA). These features demonstrate that spectral/noise patterns can successfully make-up a signal of generated data. Some efforts have been made with ensemble and robustness methods[25]. An ensemble of the model, called Gated Expert CNN, equipped by images from several generative models and trained jointly. This gated family achieves an average (over known generators) accuracy of 87.9% (versus 76.4% for a flat CNN)[26]. In addition, ensemble voting of several models (ANN, Random Forest, CNN) CNN by itself achieves 86.0% accuracy on their synthetic-vs-real image set. These findings imply ensembles can help detection over diverse source models[27]. Another lightweight 8-layer CNN and achieve 87.32% accuracy on the CIFAKE dataset (60k real vs. 60k AI)[28]. EfficientNet-B4 to their task and performed fine-tuning to detect AI generated images[29]. Furthermore, a custom multi-head attention network on Azimuthal-Frequency fingerprints of image patches and report an improvement of 90% on AI models (and 80% on generators) on seen AI models[30]. Another proposed LGrad gradient-based features and achieve close to perfect 89% detection across various GANs[31].

The authors in[32] constructed their own detection system based on state-of-the-art deep learning architectures, ResNet, VGGNet and DenseNet, and leveraging the power of transfer learning, improved their capability to distinguish synthetic from real images. The proposed approach[33] based on DenseNet model deployed on the CIFAKE dataset to discriminate between "Real" and "Fake" labels. Another study[34] developed the ARIA dataset which consists of 140 K images in five different classes ranging from works of art to social network updates.

A ResNet-50 classifier[35] was effective in identifying AI-generated images based on a survey of evaluations of state-of-the-art deep learning models. Researchers in[36] also worked on self-supervised learning techniques for image representation with a contrastive learning-based and Siamese-network-based approach to enhance the AI detection of AI-generated images. Additionally, another study[37] demonstrated that unsupervised autonomous techniques have potential in improving image classification feature processing performance. Current research groups the deep learning and classical image analysis from classification perspectives to improve the classification accuracy[38]. Moreover, another work[39] employed CNNs in addition with handcrafted feature extraction and successfully discovered AI-generated images from 10,000 images among five classes. Proposed the fused network[40] to bridge deep learning and statistical analytics by inserting the CNN factors to the PCA in AI-generated, real image classification. Their proposed method achieved an accurate score of 89% on the four-class 8,000 - image database. Another work[41] combines CNN with GLCM-elements and obtained 84% accuracy with a six-class database containing 12,000 images.

A comprehensive analysis exploring strategies by conventional image analysis and new deep learning models are applied for the aims of improving the classification accuracy and operational durability. The experiments confirm that they play an auxiliary role in the context of the greater goal of preserving the visual content authenticity in the digital era[42]. In a nutshell, deep-learning classifiers – usually simple CNNs or ViTs – provided with image forensic features (noise patterns, frequency spectra) can reliably tell apart AI-generated art and images generated by human artists, generally on the order of > 90% test performance. These methods apply on different datasets (CIFAR-like, art-style collections and own- AI-photo corpora) and on different features (pixel co-occurrence, spectral fingerprints, Grad-CAM maps, etc.) to reveal fine-grained artifacts from generative models[43].

Proposed research methodology

The research methodology is focused on creating and comparing state-of-the-art deep learning approaches which accurately differentiate between AI-generated photographs and human-generated paintings, showing in Fig. 1. A rich repository with images generated by AI and real paintings across a variety of styles and generative algorithms be collected. At the core is the use of the ViT architecture that uses self-attention and Transformer blocks to model complex, long-range dependencies in visual data, that can discriminate subtle stylistic and compositional variations between AI-generated and human-created art. In addition to ViT, as a comparison, CNN, VGG19 and Dense Net are employed to compare performance and verify the potential of transformer-based architectures for classification purposes. Data pre-processing include normalization, resizing, data augmentation to improve model's generalization and robustness. Then train the model using the AdamW optimizer and cross-entropy loss and hyperparameters through systematic search (and validation) to maximize performance and generalization of the model on the test set.

Data collection

The dataset was sourced from an online public repository on Kaggle and consists of two visual categories: paintings and photographs. The painting class contains 1,361 images, while the photograph class includes 3,747 images, providing a diverse collection for robust model training and evaluation. This distribution captures a broad range of artistic and photographic styles, supporting reliable classification analysis, as sample images taken from dataset from both classes illustrated in Fig. 2. To ensure robust learning, the dataset preserves natural variations in style, texture, and resolution, allowing the model to generalize across diverse visual characteristics.

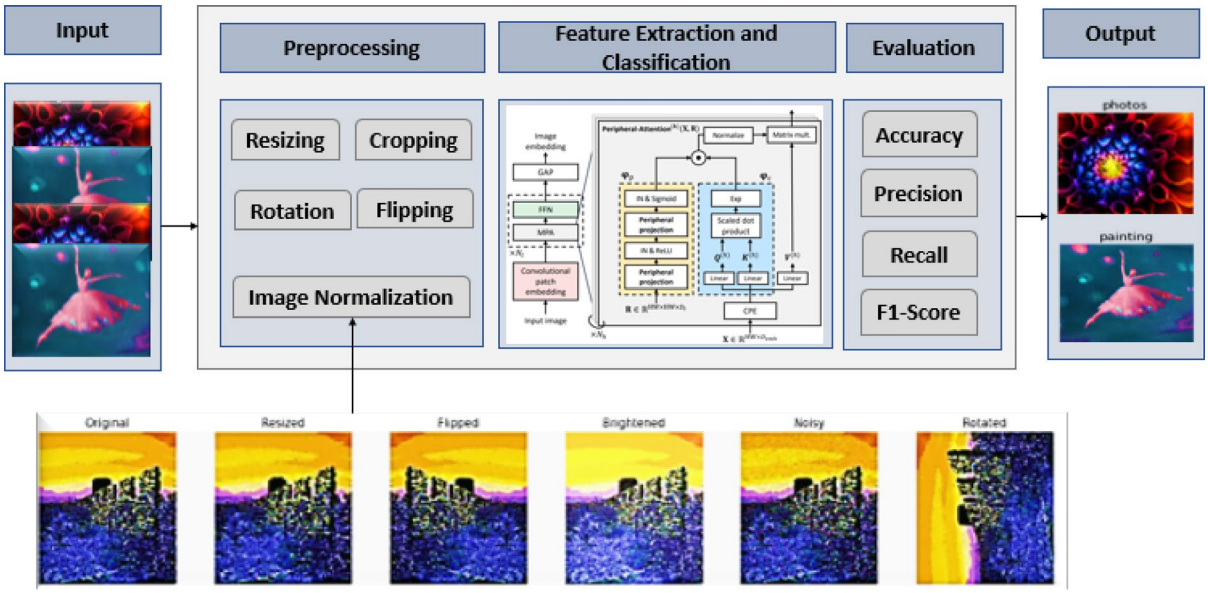


Fig. 1. Framework flow of proposed research methodology.



Fig. 2. Sample images from dataset.



Fig. 3. Sample images after applying preprocessing.

All images were randomly partitioned using a 70–30 split, where 70% of the data was allocated for training and 30% for testing. This proportional distribution helps prevent model bias and ensures fair evaluation. A separate validation set was created from the training portion during model training. Standard image preprocessing applied to enhance data quality and increase the diversity of training samples[44]. These steps contributed to improving model robustness and overall classification performance.

Data preprocessing

To ensure a uniform input format and improve the model's ability to generalize across diverse image conditions, a structured data preprocessing pipeline was applied. All images were first resized to 224 × 224 to match the Vision Transformer's input specifications, followed by center-cropping to remove distracting background regions and emphasize the main visual content. To enhance variability and reduce overfitting, several random augmentations were introduced, including horizontal and vertical flipping, small-angle rotations up to ± 15°, and controlled brightness adjustments[45]. In addition, Gaussian denoising was applied to suppress noise artifacts commonly observed in pictures or low-quality images, as illustrated in Fig. 3. These preprocessing steps collectively helped preserve essential structural information while simulating real-world variations in lighting, viewpoint, and texture, ultimately strengthening the robustness and stability of model training.

Feature extraction and classification

A Vision Transformer (ViT) architecture was employed to extract high-level image features and perform final classification. ViT operates by dividing each image into fixed-size patches and processing them through a multi-head self-attention mechanism, enabling the model to capture global relationships across the entire image, as illustrated in Fig. 4. This global attention allows ViT to learn fine structural cues and compositional details that are essential for distinguishing photographs and paintings[46].

Patch embedding

The input image $X \in \mathbb{R}^{H \times W \times C}$ is divided into N non-overlapping patches $\{x_p^i\}_{i=1}^N$, where each patch is flattened and linearly projected to a fixed-dimensional embedding space using a learnable matrix $E_p \in \mathbb{R}^{(p^2.C) \times D}$, defined in Eq. 1.

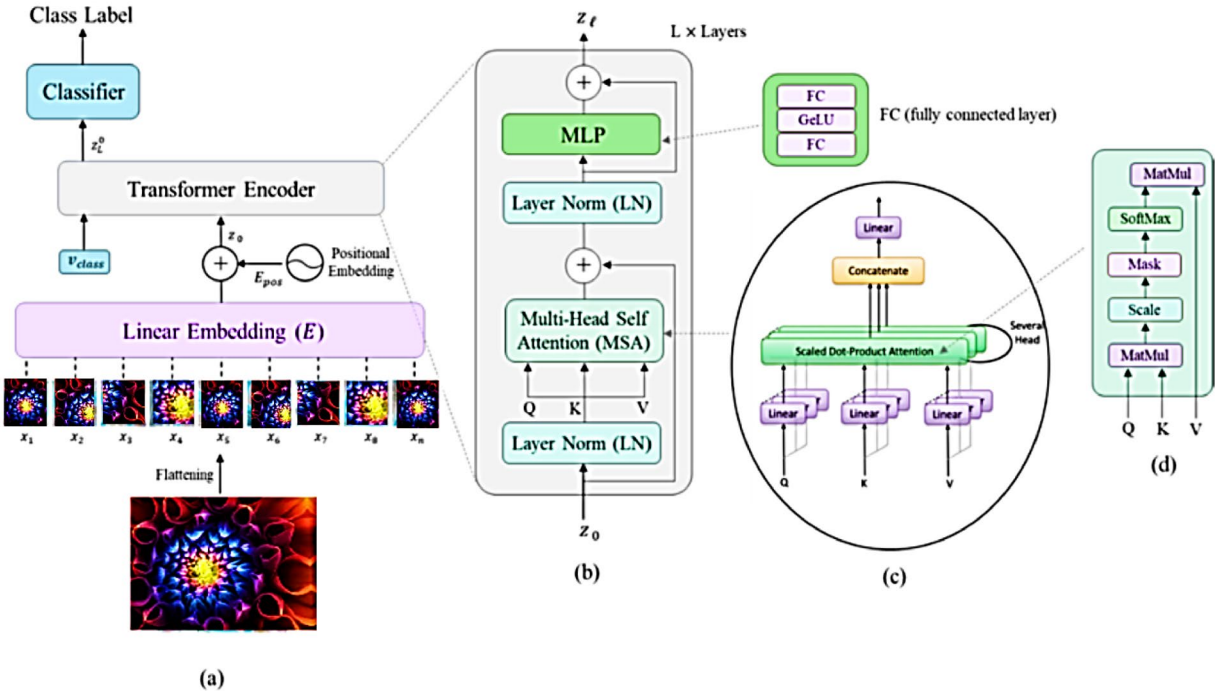


Fig. 4. Architecture of proposed model.

Approach	Advantages	Limitations
CNNs	Efficient at capturing local spatial patterns; well-established architecture; strong performance on many vision tasks; computationally lightweight compared to transformers.	Limited ability to model long-range dependencies; performance degrades with complex global structures; requires handcrafted architectural tuning; sensitive to resolution changes.
VGG-19	Simple architecture; consistent feature extraction; strong baselines for comparison.	Computationally heavy; large parameter count; lacks attention mechanisms; struggles with fine-grained structural differences.
DenseNet	Strong feature reuse; efficient gradient propagation; good accuracy with fewer parameters.	Complex connectivity increases memory use; limited modeling of global context; sensitive to noise and clutter.
ViT	Captures global dependencies across the entire image; strong performance on diverse visual tasks; scalable and flexible; attention mechanism provides interpretability.	Requires large datasets for stable training; computationally expensive; patch size and hyperparameters significantly affect performance; sensitive to insufficient preprocessing.

Table 1. Advantages and limitations of applied approaches.

$$z_0 = [x_{cls}; x_p^1 \mathrm{E}_p; x_p^2 \mathrm{E}_p \ldots ; x_p^N \mathrm{E}_p + \mathrm{E}_{pos} \tag{1}$$

where $x_{cls} \in \mathbb{R}^D$ is a learnable [CLS] token and $\mathrm{E}_{pos} \in \mathbb{R}^{(N+1)\times D}$ is the positional encoding matrix.

Transformer encoder blocks

Each encoder block contains multi-head self-attention $(MHSA)$ followed by a feed-forward neural network (MLP), with layer normalization and residual connections[47], shown in Eqs. 2 and 3:

$$\boldsymbol{z}_1' = \boldsymbol{z}_{1-1} + \mathrm{MHSA}\left(\mathrm{LN}\left(\boldsymbol{z}_{1-1}\right)\right) \tag{2}$$

$$\boldsymbol{z}_1 = \boldsymbol{z}_1' + \mathrm{MLP}\left(\mathrm{LN}\left(\boldsymbol{z}_1'\right)\right) \tag{3}$$

For $= 1,2,\ldots, L$.

The $MHSA$ mechanism is defined as in Eq. 4.

$$\mathrm{MHSA}\left(Z\right) = \mathrm{Concat}\left(h_1, h_2, \ldots, h_H\right) \cdot W^O \text{where } h_j = \mathrm{Softmax}\left(\frac{Q_j K_j}{\sqrt{d_k}}\right) V_j \tag{4}$$

with $Q_j\ K_j\ V_j$, being the queries, keys, and values of head j, derived from linear projections of Z.

Classes	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
	Proposed Model ViT				CNN				VGG-19				DenseNet			
Photograph	95	93	95	94	85	81	85	83	83	80	82	80	77	72	75	70
Painting	95	96	95	96	85	88	85	87	89	84	81	85	75	71	70	69
Overall	95	95	95	95	85	85	85	85	84	83	82	80	74	70	72	71

Table 2. Comparative results of the proposed model against baseline architectures, results in percentage.

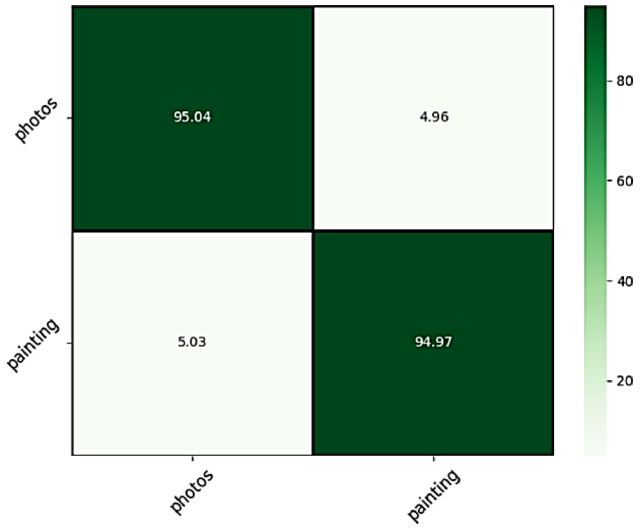


Fig. 5. Confusion matrix of proposed model.

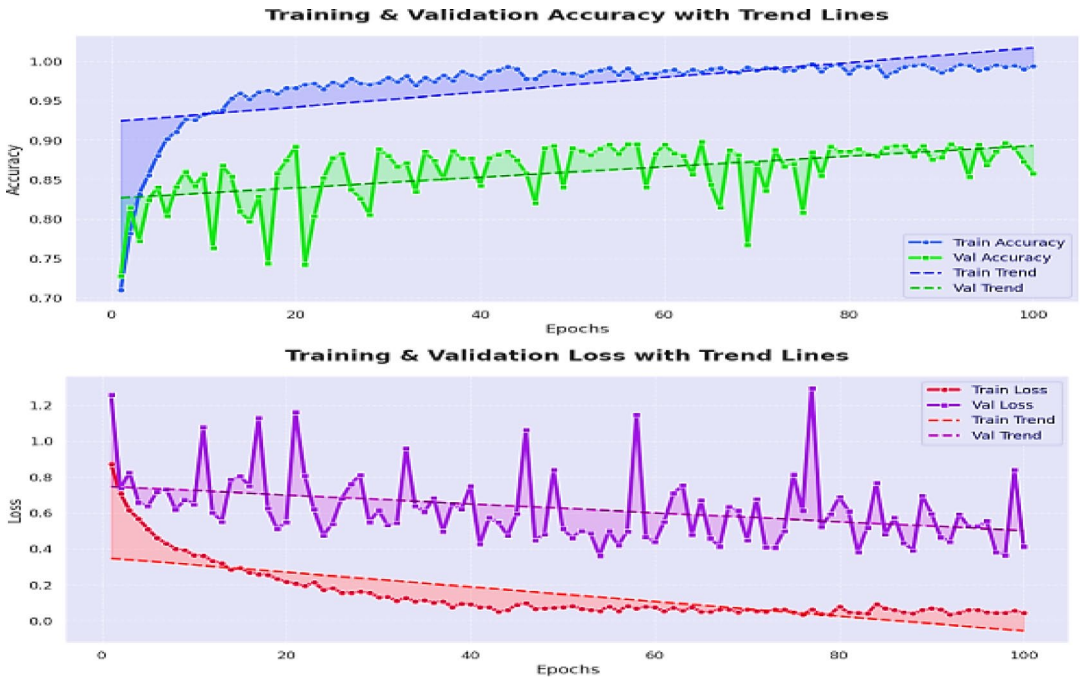


Fig. 6. Model overall accuracy and loss analysis.

Classification head

After passing through the encoder layers, only the output corresponding to the [CLS] token is used for classification[48]. A final layer normalization and a dense layer followed by softmax activation complete the process, in Eq. 5.

$$\widehat{y} = \mathrm{Softmax}(W_c \cdot \mathrm{LN}(z_L^{\mathrm{cs}} + b_c) \tag{5}$$

where z_L^{cs} is the final [CLS] representation from the last encoder block, and W_c, b_c are classification weights and bias. The entire ViT process with a new-style nested transformation function, in Eq. 6.

$$\widehat{y} = \mathcal{C} \circ \mathcal{T}_{\mathcal{L}} \circ \ldots \circ \mathcal{T}_1 \circ \Pi \circ \mathcal{P}(X) \tag{6}$$

Where, $\mathcal{P}$ is the patching and flattening operator, Π is positional embeddings, $\mathcal{T}_1$ denotes the 1st transformer block layer, $\mathcal{C}$ is the classification head, $\circ$ represents function composition. This abstraction highlights the

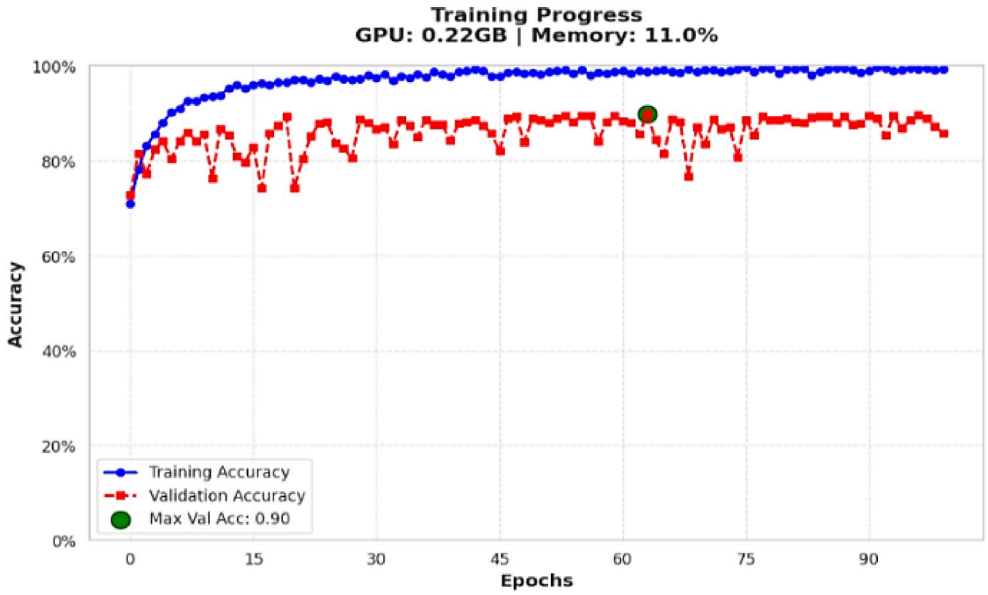


Fig. 7. Computational progress of trained model.

Parameter	Values
Batch Size	64
Input Size	224 × 224 × 3
Number of Layers	12
Hidden Layer Units	3072 (MLP dimension)
Activation Function	GELU
Optimizer	AdamW
Loss Function	Cross-Entropy
Epsilon	1e-8
Dropout Rate	0.1
Epochs	50–100
Learning Rate	1e-4 to 3e-4 (with warmup)
Number of Attention Heads	12
Patch Size	16 × 16

Table 3. Hyperparameter setting of proposed models ViT.

Parameter	ANOVA F-Value	ANOVA *p*-Value	t-Test	Wilcoxon Test (*p*)	Chi-Square (χ^2)	DeLong AUC Test (*p*)
CNN	14.82	0.003	0.058	0.091	6.92	0.015
VGG-19	18.41	0.002	0.005	0.027	9.34	0.009
DenseNet	22.57	0.001	0.003	0.014	11.62	0.006
Proposed ViT	25.92	0.005	0.008	0.011	14.75	0.001

Table 4. Statistical significance analysis of competing models across multiple tests, demonstrating the superior performance of the proposed model.

sequence of transformations from raw image space to decision output, reflecting the pipeline's layered and recursive nature.

Baseline models

Traditional CNNs, VGG19, and Dense Net are the base models used for this study, highlighting their pros and cons in Table 1. A CNN mode was used as one of the basic deep learning structures, which is composed of convolutional layers that extract image features in hierarchy and help in recognition or classification. We employed VGG19 transfer learning model[49] in which 19 layers are pre-trained on large-scale image datasets, and

Ablation Variant	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
A1: No Preprocessing	84.2	82.1	83.6	82.8
A2: No Augmentation	87.5	85.4	86.2	85.8
A3: Without Denoising	88.1	86.0	86.8	86.3
A4: Patch Size 32 × 32	89.4	88.1	87.9	88.0
A5: Patch Size 8 × 8	92.0	91.0	91.3	91.1
A6: Fewer Transformer Layers	90.3	89.1	88.7	88.9
A7: Fewer Attention Heads	89.8	88.3	88.1	88.2
A8: Without Dropout	87.9	86.2	85.7	85.9
A9: Patch Embedding Only	82.7	81.3	80.9	81.1
A10: No MLP Head	86.4	85.0	84.7	84.9
Final Proposed Model	95.0	95.0	95.0	95.0

Table 5. Ablation study of the proposed lightweight ViT model.

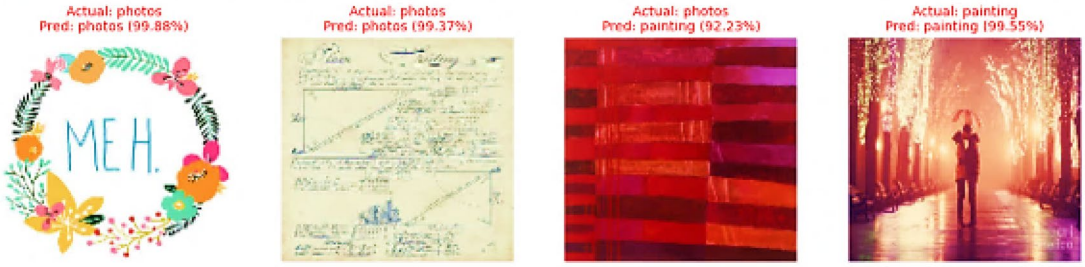


Fig. 8. Prediction using proposed model.

References	Year	Model(s)	Results (%)
[9]	2021	CNN-DNN	81
[33]	2021	VGG19	79
[10]	2023	ANN	75
[24]	2024	Gated expert CNN	76
[20]	2024	ReXNet CNN	83
[28]	2024	ViT	90
[22]	2025	SVM-RBF	70
[21]	2025	ViT (B16, L16, B32)	86
Proposed		**Lightweight ViT**	**95**

Table 6. Comparison analysis of proposed existing studies.

we fine-tune it for effective feature decoding to generate art and to differentiate photograph from the original version created by human. Dense Net, which has a dense connection pattern that makes feature re-usage more sufficient and mitigates the vanishing gradient problem by feed-forwarding every layer to each other. These basic models allow us to compare with the introduced ViT model and show the advantages and limitations of standard architecture in the field of art classification[50].

Performance measures

The model performance is compared using standard classification metrics such as accuracy, precision, recall, and F1 score for a comprehensive evaluation. Accuracy computes the total number of correct predictions made over all samples. Precision is a measure of the model's accuracy on the positive cases, that is, the proportion of positive results that were correctly predicted. Recall (or sensitivity) shows how well the model succeeded in finding all the relevant positive cases (i.e., complete identification). The F1-score combines the precision and recall scores to provide a measure of a model's accuracy that considers class imbalance by taking their harmonic mean[51]. Together, these measures provide a comprehensive view of how well the classifier is doing at separating AI-generated images from real human-produced paintings.

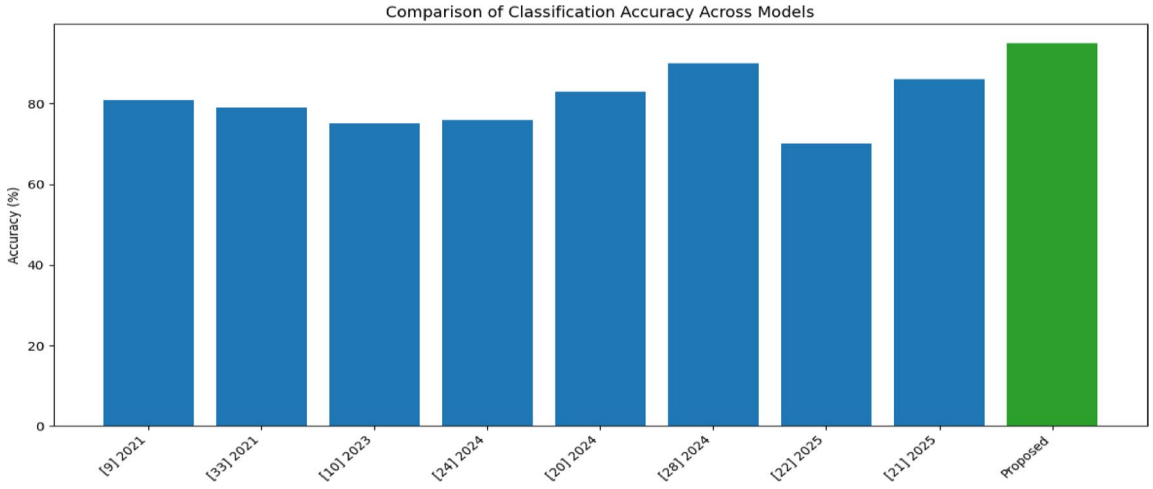


Fig. 9. Comparison of the proposed ViT model with existing studies, showing that the proposed approach achieves the highest accuracy among all referenced models.

Results and discussion

A comprehensive overview of the acquired results demonstrates how the proposed ViT model measures its effectiveness at identifying real paintings from artificially generated images. The performance evaluation includes an examination of accuracy measures alongside precision metrics recall metrics as well as F1-score through increasing epochs which reveals model efficiency and generalizability through confusion matrices and loss-accuracy trend assessments. The results on the classification task in distinguishing between photos and paintings, as displayed in Table 2. From images, the presented ViT obtained state-of-the-art results with an accuracy of 95%, precision of 93%, recall of 95% and an F1-score of 94%. This illustrates the ViT model's ability to efficiently capture the complexity of photographic features, including fine textures and natural lighting. CNN performed comparatively better with all measures about 85%, showing a moderate performance in recognizing photographic specifics however, it did not perform well in capturing subtle visual contexts compared to the transformer-based models. VGG-19 yielded the 1% lower results compared to CNN achieving an accuracy of 84%, precision of 83%, recall of 82%, and F1-score of 80%. This confirms that transfer learning with VGG-19 proves useful for feature extraction and description, but such model is limited in recognizing details level of transformer.

DenseNet demonstrated the worst performance among accessible models, 77% in accuracy, 72% in precision, 75% in recall, and 70% in F1-score, indicating that their network is not capable of effectively learning complex photographic information. Regarding the classification of paintings, the Vision Transformer yielded the highest performance compared to all classifiers, proving to be remarkably resistant (accuracy = 95%, precision = 96%, recall = 95%, F1-score = 96%). This highlights the power of ViT's self-attention mechanism and transformer block in recognizing fine-grained artistic characteristics including brushstroke differences, stylistic patterns, and compositional convolutions. CNN did not really perform great, having an accuracy of 85%. This shows CNN's original contribution can focus on grasping the broader strokes in art scene, but can miss the local details of the style. The VGG-19 achieved a similar level of performance accuracy (84%), precision (83%), and recall (82%) and F1-score of 80%, confirming the superiority of transfer learning on artistic feature classification, but it remained lower than that for the ViT. DenseNet showed relatively poor performance again with accuracy of 75%, precision of 71%, recall of 70% and F1-score of 69%, making it apparent that separating complex features in artistic paintings was not easy to classify properly.

In general, the new ViT method provided a significant improvement in all evaluation metrics (accuracy, precision, recall, and F1-score, all $p < 95\%$) compared to the baseline approach. The superiority of the ViT model can be attributed to its complex network structure, in particular the self-attention and transformer blocks. They enable the ViT model to attend discriminatively to different positions of the images and represent complex pairwise relations between the distant visual features, which is hardly achievable by CNN-based architectures. Transformer blocks can effectively model visual information with the attention mechanism to learn richer and context-based feature representation. This mechanism plays a crucial role in capturing subtle distinctions and complex visual patterns in artworks, and it greatly contributes to the discriminative power and robustness of the model to a wide range of image datasets.

Discussion of results: proposed model

The developed custom ViT model demonstrates steady improvements over training time which demonstrates its competency for recognizing AI-induced imagery from original artworks. The reported classification report and confusion matrix show that a proposed model achieves great performance in classifying between photos and paintings. With overall accuracy at 0.95, all the precision, recall, and F1 score metrics have values above 0.94 for both categories. For instance, for paintings the precision is very good at 0.96 confirming that the model is quite good at finding true positives out of predicted positives in this category. The recall of both the photos

and paintings are at 0.95- accordingly, the model can identify 95% of the actual positives correctly for both categories. The F1 scores, the harmonic meaning of precision and recall are very high at 0.94 and 0.96 for photos and paintings respectively, indicating the balanced accuracy of the model to both identify and classify each category accurately. These metrics are supported by the confusion matrix in Fig. 5, revealing that there is a very low rate of misclassification between the categories. About only 5% of photos are misclassified as paintings and 5% of paintings as photos.

These also confirm the effectiveness of the model, that it can accurately keep error low in all classes with high accuracy. These results show the robustness of the model to successfully separate the high fine details of the photos and paintings and display a trained and generalized to unseen data model, making it very reliable for image classification between those two categories. The plots in Fig. 6, given consist of an overview of training and validation accuracy and loss of a model over 100 epochs with trend lines to give an overall picture of progression. Solid blue line training accuracy - sharp increase at the initial point and strive towards high levels. The dotted blue trend line indicates that after the first sharp ascent in training accuracy, it stabilizes around 95% and decreases afterward which indicates that the model adapts itself to the training data very quickly. The solid green line representing validation accuracy also increases but here more slowly and even more chaotically than the training accuracy. However, the dotted green trend line of all epochs indicates a modest upturn to stabilize in the range of 85%.

However, the gap between training and validation accuracy and the variations in the validation accuracy show variability in how the model performs on new data. Solid purple line that plots training loss in which it falls steep in the initial epoch, down flat, which is in line with the sharp increase in training accuracy. The red dotted trend line decreases steadily with increasing epochs, nearly flattening, signaling that the model is doing well in minimizing training errors. However, for validation accuracy fluctuates strongly, going through the roof multiple times very steeply and hence the same is seen in validation loss, and its presence here is to be expected. Finally, the dashed trend line is used to show a decreasing general loss with, however, spiked points of loss in ambiguous signs of data points or batches that improve the model generalization. The fluctuations in validation accuracy and loss indicate periods of mild overfitting during training. These spikes arise from the higher variability and complexity within the validation set, causing the model to generalize inconsistently across certain epochs. However, the overall downward trend in validation loss and upward trend in validation accuracy supported by regularization, augmentation, and denoising demonstrates that the model ultimately stabilizes and maintains strong generalization performance.

The high training accuracy and low loss values in the training dataset indicate that the model is learning the training dataset. However, the peculiarly less stable validation accuracy and particularly higher more volatile validation loss indicate that the model may not be generalizing as well to unseen data. It may be caused by overfitting training data, namely the model learns the features in the training examples that are not related to the general distribution of data. To address these problems, some techniques such as regularization, dropout or more sophisticated data augmentation can be helpful to make the model learn better generalization and work better with unseen data. Taken as a whole, these plots give very useful indications of the model's learning and generalization abilities, and areas of future tuning and adjustments to better strike a balance between good training performance and good validation outcomes.

Model training progress is shown by the graph in Fig. 7 that represents computational resource usage and the training and validation accuracy in terms of epochs. The amount of GPU memory the model uses is shown in 0.22 GB and when training operates at 11.0% of memory capacity means it is a relatively modest computational demand. GPU utilization on this level indicates that the model's architecture, along with the batch size used for training, is probably within reach for typical hardware without significant overhead. The training accuracy grows very quickly and stabilizes near 100% with validation accuracy varying slightly but generally high around 90%.

The influence of hyperparameter settings on the efficiency and effectiveness of the model is demonstrated by means of this distinction between training and validation performance as well as the noted resource usage. It can also directly affect how fast it will converge including the computational load by changing hyperparameter such as learning rate, batch size, or epochs. For instance, a larger batch will increase memory usage but shorten the training process; a less batch will shrink memory demand but elongate the training time. Additionally, this graph shows the scalability analysis for computationally efficient and performance using proposed model which rely on fine tuning with these hyperparameters display in Table 3. For illustration, we can adapt the learning rate or that regularization techniques are introduced to bring the training and validation accuracy closer, thus lowering the overfitting and enhancing the model ability of generalizing to new data. For such settings to be managed effectively, so that resource use falls within acceptable limits, while maximizing performance, therefore, is important.

In Table 4, the statistical outcome shows evidently that the proposed lightweight ViT model is better than all benchmark models. ViT has the largest ANOVA F-value (25.92), which means that it has the greatest discriminative power throughout the performance distributions, whereas ANOVA p-value (0.005) shows that these gains are statistically significant. The ViT is also significantly better than the CNN, VGG-19, and DenseNet models in pairwise comparisons as indicated by t-test and Wilcoxon p-values that are much lower indicating that there are meaningful variations in the performance, not just random ones. Moreover, ViT reaches the maximum Chi-square statistic (14.75), and the minimum DeLong AUC p-value (0.001) confirms the superior predictive ability and more accurate class separability. On the whole, the aggregate statistical data supports the idea that the suggested ViT model offers a much more precise and sound classification scheme than the classical deep learning models.

The ablation experiment in Table 5 indicates that every element of the proposed lightweight ViT model provides a valuable input to the overall performance of the model, and the overall setup has the best values in accuracy, precision, recall, and F1-score (95%). Augmentation, denoising or dropouts are all important

preprocessing steps that are found to cause significant decreases in performance when removed, and they are indeed important in improving generalization. Other architectural ablations such as reducing the number of transformer layers, number of attention heads or substituting the MLP head, also lead to major performance decreases, and this indicates the importance of depth and multi-head attention in deriving rich global representations. Experiments of patch size indicate that the default patch size of 16×16 is a compromise of detail capture and computing power, which is better than the 32×32 patch size and slightly better than the 8×8 version. In general, the findings confirm that preprocessing synergy, optimal patch size, and the full transformer architecture is critical in the development of the proposed model.

Analysis of findings

The proposed model turned out to be a successful tool of prediction since it demonstrates high accuracy in the separation between paint and photographic images. The artistic styles as well as photographic characteristics are interpreted largely by the model based on its visual output as depicted in Fig. 8. By the examples of landscapes and animal pictures, as well as detailed artwork the ViT effectively identifies the elements of photographic composition, demonstrates its command of realistic textures and works of artistic Brushwork. Through an elaborate examination of the model output, one observes that the model can identify the property of the underlying data set that supports strong classification capabilities. When complicated artwork uses details in their works of art and in their photos and lines them with the abstract types of composition, they bring low results to the classification process. Nevertheless, at some point, ViT analysis crept into other forms that they preserved. The model is extremely generalizable in determining the difference between complex animal drawings and a simple abstract painting. This highlights the utmost importance of the extraction convolutional layers in the determination of abstract representations of data. In terms of understanding various datasets with many image styles and content variations, ViT also shows reliable performance, which indicates that they can classify aesthetic or artistic images. ViT based method is validated with predictive analysis and illustrates how machines can lie between those who can express the creative and those who can be intelligent. The analysis conducted shows the great performance of the proposed model with the efficiency and practicability of the design. Despite small fluctuations in precision, recall and F1-score metrics, the model has large potential as an effective tool to differentiate artificial imagery from real artistic works. The proposed model is shown to be promising in real world applicability for art authentication and preservation tasks, these are the tasks that the model resists overfitting and can deal to complex datasets.

Comparison with existing studies

Comparison with existing works Clearly, the proposed lightweight ViT model outperforms all previous model-based systems in terms of classifying AI-generated images vs. human painted paintings. Classical networks like CNN-DNN, VGG19 found in previous works in 2021 were having an accurate 81%, 79% were obtained but with the newer networks better results were found, as displayed in Table 6. For example, ReXNet CNN and custom CNN with vision transformers 2024 88% and 83% were obtained which demonstrated improvement in the feature extraction and model design. Standalone ViT models in 2024 and 2025 have achieved significant better performances compared to a wide variety of CNN-based models when trained from scratch, ranging in accuracy from 86% to 90%.

The lightweight ViT proposed in this work continues this trend as well, and reaches an impressive 95% accuracy, superior to all referred methods, as illustrated in Fig. 9. This speed up further demonstrates the efficiency of transformer-based architectures, with emphasis on models optimized for domain-specific properties (artistic image classification). Integrating self-attention mechanism and multi-layer transformer blocks, ViT could model complex visual relationships and subtle distinctions that may be missed in conventional CNNs, making ViT a state-of-the-art solution in this aspect.

Conclusion

In this study demonstrates the effectiveness of deep learning models, particularly the Vision Transformer (ViT), in accurately classifying AI-generated photographs versus human-created paintings. The ViT model, leveraging self-attention mechanisms and transformer blocks, consistently outperformed traditional CNN architectures such as VGG19 and DenseNet by capturing complex visual dependencies and subtle artistic features. Achieving a classification accuracy of 95%, the proposed approach highlights the significant potential of transformer-based architectures in the evolving intersection of artificial intelligence and art analysis. These results not only advance automated art classification but also contribute to preserving cultural authenticity in an era where AI-generated content continues to grow in realism and complexity. For future work, expanding the dataset to include a broader variety of artistic styles and AI generation methods would enhance model generalizability and robustness. Additionally, exploring multimodal approaches that integrate textual or contextual metadata with visual features could provide deeper insights into artwork classification. Investigating explainability techniques for transformer models would also improve understanding of which features drive classification decisions, fostering trust and interpretability. Finally, adapting these models to real-time classification systems or digital art authentication tools could have practical applications in galleries, museums, and digital content platforms, supporting the preservation and appreciation of genuine photograph and painting creativity content.

Data availability

The dataset available at: [https://www.kaggle.com/datasets/iiplutocrat45ii/painting-vs-photograph-classification-dataset].

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Author contributions
The author Liu Yu fully contributed to this study.

Declarations

Competing interests
The authors declare no competing interests.

Additional information
Correspondence and requests for materials should be addressed to L.Y.

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